

NUMERICAL METHODS AND OPTIMIZATION

NMO – 3rd Year

ENSIA — Spring 2026

LAB 1

- Number Representation and Accuracy -

Objectives

- Getting started with *Octave*
- Perform calculations on numbers
- Manipulate variables and functions
- Understand the effect of a machine's properties on calculations.

Exercise 1:

- **Q1:** Specify the value of the variable 'a' after the following command: 'a = int8(400)'
- **Q2:** Specify the value of the variable 'a' after the following command: 'a = uint8(-15)'
- **Q3:** Specify the value of the variable 'a' after the following command: 'a = uint8(105.75)'
- **Q4:** Giving only 2 digits after the decimal point, specify the value of the variable 'a' after the following command: 'a = pi + 2':
- **Q5:** Specify the value of the variable 'a' after the following commands:"

```
>> pi = 10;  
>> a = pi+2;
```

- **Q6 :** Type the following commands

```
Fenêtre de commandes  
>> x = 70000;  
>> y = 7e4;  
>> z = 0.07e6;  
>> t = 700000e-1;
```

What do you observe:

- All variables are integers.
- All variables contain the same value.
- y is in scientific notation.
- The ";" prevents the result from being displayed directly.
- Variables y, z, and t are complex.
- Variables are internally represented in 32-bit floating-point (single precision).

- **Q7:** At the end of the following commands, the display of *y* shows a strange value: 'Inf'. What does this mean?

```
Fenêtre de commandes
>> x = 14e128;
>> y = single(x);
>> disp(y)
Inf
```

- **Q8:** in the commands below we observe that the variable 'a' does not contain exactly the value it should (350), why ?

```
Fenêtre de commandes
>> x = 350;
>> a = int8(x);
>> b = int16(x);
>> c = int32(x);
>> d = int64(x);
>> disp(a)
127
>> disp(b)
350
>> disp(c)
350
>> disp(d)
350
```

Exercise 2:

- **Q1 :** Type the commands: *realmin* and *realmax*, what does it mean the output of each command?
- **Q2 :** Calculate the product of **a** and **b** such that : $a = 0.56 \times 10^{202}$ and $b = 1.2 \times 10^{107}$, explain the result.
- **Q3 :** Calculate the sum of **c** and **d** such that : $c = 0.789 \times 10^{-76}$ and $d = 1.23 \times 10^{-350}$, explain the result.
- **Q4 :** Define the numbers $x = 1.23004 \times 10^{-4}$ and $y = -1.23003 \times 10^{-4}$ calculate $x+y$ in bank, short, and long format, what do you observe ?
- **Q5 :** Type the command *eps*, what does it represent ?
Suppose that we are trying to accurately evaluate the following function for small values of *x* on the computer:

$$f(x) = \frac{1}{1 - \sqrt{1 - x^2}}$$

We define the function f on *Octave* as follows :

```
>> f=@(x) (1./(1-sqrt(1-x.^2)));
```

- Write the command to evaluate the function at $x = \frac{\sqrt{\epsilon_M}}{2}$, explain the result.
- Propose another function g to avoid this error.
- Evaluate the function g at $x = \frac{\sqrt{\epsilon_M}}{2}$